BlockCloud: Developing Blockchain Solutions in the Cloud

BlockCloud is an upcoming 'blockchain based TCP/IP architecture' that brings together blockchain technology with service-centric networking to support Internet of Things (IoT) oriented solutions. Developed by a group of Chinese academicians, it aims to focus on creating highly mobile blockchain solutions that are not restricted by heavy blockchain platforms. It can also operate as a layered solution to solve all sorts of scalability issues faced by almost every blockchain platform these days.

According to a white paper by the BlockCloud team, more than 22 million IoT devices will be connected to the Internet by the year 2022. The influence of IoT will be seen across multiple sectors like healthcare, augmented reality, autonomous vehicles, smart cities, grids, etc. However, along with this will come the problem of poor IoT architecture. To address this challenge, BlockCloud teams aim to solve issues like connectivity failure, scalability, trust and privacy, security and flexibility.

BlockCloud technology is considered an extension of the current Internet to transform it into the next generation Internet. It combines blockchain technology and next-generation Internet technology to build an underlying network protocol in terms of 'building blocks', to support seamless connectivity for dynamic networks and the interconnection of upper layer applications in a secure and efficient manner.

BlockCloud introduces service-centric networking (SCN), through which the service access layer (SAL), control plane and service plane are separated. Without any use of complex IP addresses, the upper application layer can connect to the service using the service name, leading to improvement in mobility and scalability. In addition, SCN networks improve the security level via blockchain technology.

A proof of service (PoS) mechanism verifies all services, designs the distributed peer-to-peer network to save service information, and implements a truthful continuous double auction (TCDA) mechanism to distribute services. In addition, the CoDAG (compacted directed acyclic graph) structure effectively records the transactions. The hybrid consensus protocol combines both permission and permissionless protocols to achieve outstanding results.

Focusing on a new PoS consensus, BlockCloud tries to resolve blockchain transactions simply by having the relevant devices share information between themselves. This allows for a clean verification environment, with unsavoury participants being penalised and restricted from using the network.

As a multi-layered solution, BlockCloud introduces a few new concepts.

- *Edge computing:* This enhances current connectivity levels between different computing devices
- *Internet of Vehicles:* A communications solution for smart vehicles
- *Smart home:* Blockchain-enabled functionality to power smart homes of the future
- *Smart health:* A system for improved delivery of various health-oriented services using IoT devices